

ASSIGNMENT TASK

Assignment Title

Design, Assess and Validate a Secure Smart Manufacturing Environment Using Ethical Hacking Techniques

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ITA1408 – Ethical Hacking

Mapped Course Outcomes

- **CO3: Analyze network services using port scanning, ping sweeps and enumeration techniques across operating systems to identify vulnerabilities.**
- **CO4: Implement network security testing methods by assessing web-server, web-application and wireless-network security in an authorized Kali Linux laboratory.**
- **CO5: Analyze vulnerabilities in Windows and Linux systems and evaluate security weaknesses using appropriate ethical-hacking tools and techniques.**

Bloom's Taxonomy

L6 – Create

SDG Mapping

SDG 9 – Industry, Innovation and Infrastructure

Actual Assignment Problem

You are required to design, simulate, secure and validate a cybersecurity architecture for a smart healthcare centre containing:

- Patient-monitoring devices
- Wearable sensors
- Clinical workstations
- Windows servers
- Linux servers
- Wireless IoT network
- Electronic Health Record (EHR) portal
- Remote healthcare access

The assignment requires you to use the CICIoMT2024 dataset to analyze normal and malicious network traffic.

Security testing must be performed only in an authorized, isolated virtual laboratory using tools such as:

- Nmap
- Wireshark
- Nikto
- OWASP ZAP
- Python
- Kali Linux utilities

The assignment requires identification and prioritization of:

- Network vulnerabilities
- Web vulnerabilities
- Wireless vulnerabilities
- Windows vulnerabilities
- Linux vulnerabilities

You must then design a secure architecture containing:

- Network segmentation
- Firewall
- Access-control rules
- Multi-factor authentication
- Role-based access control
- Encryption
- Secure wireless communication
- Intrusion detection
- Centralized logging
- Patch management
- Backup and recovery controls

Finally, the proposed architecture must be validated using measurable parameters such as:

- Exposed services
- Attack-detection rate
- False alarms
- Response time
- Network availability
- Resource utilization

2.) What You Actually Need to Submit

1. Problem Statement and Problem Formulation

Explain the security problem of a smart healthcare environment and formulate the need for a secure architecture.

2. Objectives

For example:

1. Design a secure smart healthcare network architecture.
2. Analyze IoMT traffic using CICIoMT2024.
3. Identify normal and malicious traffic.
4. Perform authorized network reconnaissance.
5. Identify exposed network services.
6. Assess Windows and Linux security.
7. Assess web-server and web-application security.
8. Assess wireless-network security in an isolated laboratory.
9. Implement appropriate security controls.
10. Validate the architecture using measurable security metrics.

3. Requirements, Constraints and Assumptions

Hardware

- Laptop/PC
- Minimum 8 GB RAM recommended
- Virtualization support

Software

- Kali Linux
- Windows virtual machine
- Linux virtual machine
- Wireshark
- Nmap
- Nikto
- OWASP ZAP
- Python
- VirtualBox/VMware
- Git/GitHub

Dataset

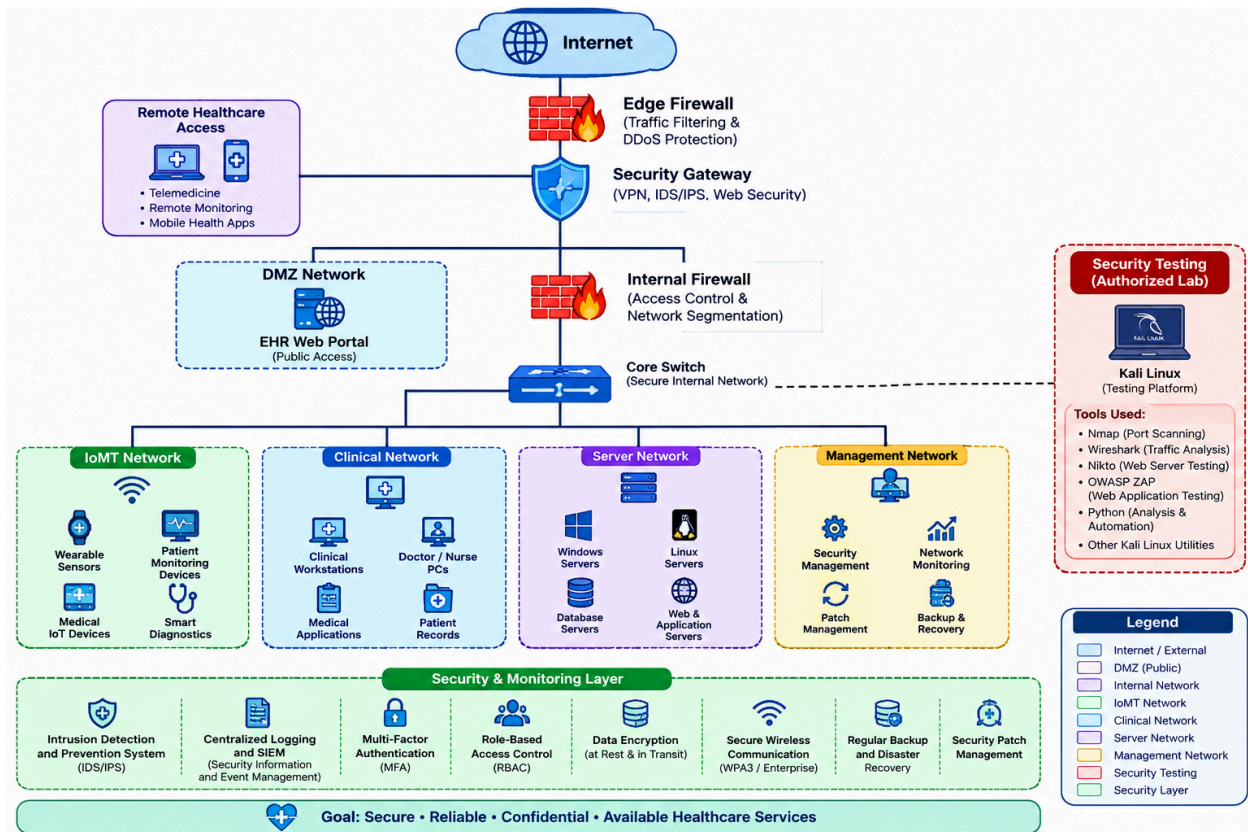
CICIoMT2024

The dataset is used for analyzing IoMT network traffic and distinguishing normal and malicious behavior.

Important Constraint

All scanning and testing must be performed against authorized laboratory systems only.

4. Proposed Architecture



5. Modules

I recommend dividing your project into **6 modules**.

Module 1 – Smart Healthcare Network Design

Design the complete network architecture containing:

- IoMT devices
- Wearable sensors
- Clinical systems
- Windows servers
- Linux servers
- EHR portal
- Wireless network
- Remote access
- Firewall
- IDS
- Central logging

Module 2 – Dataset Analysis

Use **CICIoMT2024** to analyze:

- Normal traffic
- Malicious traffic
- Protocols
- Source/destination information
- Traffic patterns
- Attack categories

Generate graphs and statistical observations.

Module 3 – Network Security Assessment

Use authorized Kali Linux tools:

nmap

for port/service discovery and enumeration.

Use:

Wireshark

for packet and traffic analysis.

The goal is to identify exposed services and potential security weaknesses.

Module 4 – Web and Wireless Security Assessment

Assess the authorized laboratory web environment using:

Nikto

OWASP ZAP

For wireless testing, use only an **isolated lab wireless network that you own or have explicit permission to test.**

Module 5 – Windows and Linux Security Assessment

Analyze security weaknesses in the laboratory Windows and Linux systems.

Check areas such as:

- Open services
- Unnecessary ports
- Weak configurations
- Missing patches
- Authentication controls
- Access permissions
- Logging
- Security policies

Module 6 – Security Controls and Validation

Implement controls such as:

Firewall

+

Network Segmentation

+

MFA

+

RBAC

+

Encryption

+

Secure Wi-Fi

+

IDS

+

Centralized Logging

+

Patch Management

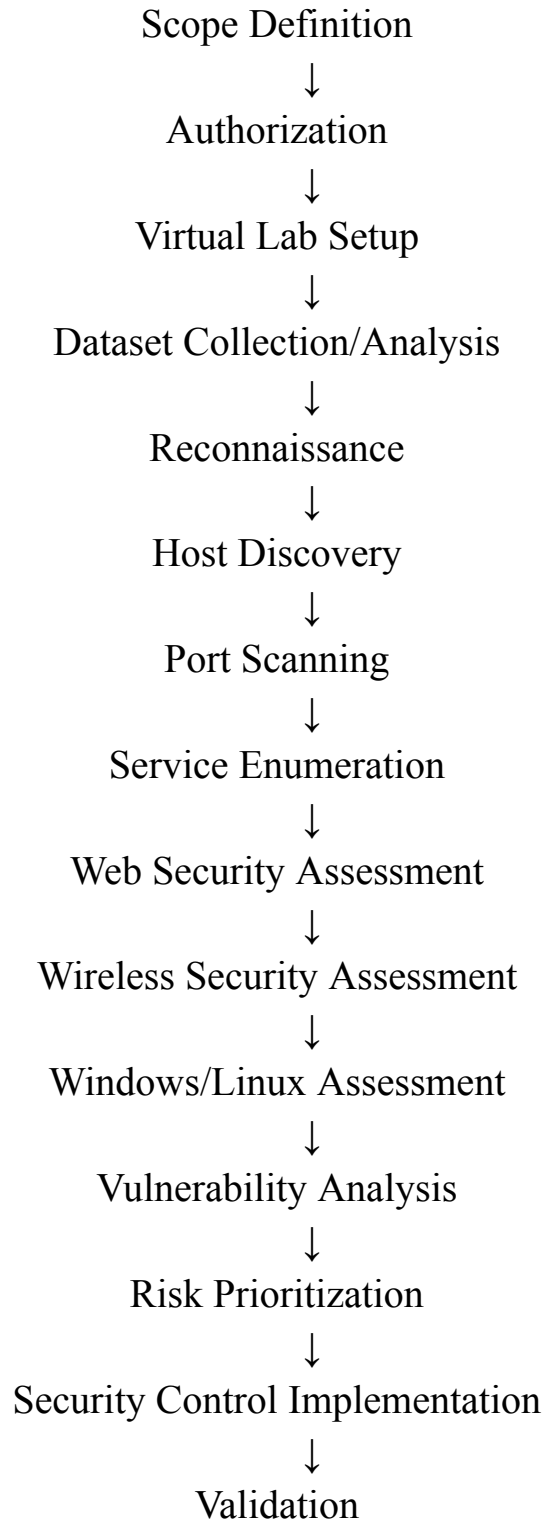
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Backup & Recovery

Then measure the improvement.

6. Ethical Hacking Methodology

Your methodology can be:



7. Kali Linux Tools

Your report can include a tool-purpose table:

Tool	Purpose
Nmap	Host, port and service discovery
Wireshark	Network packet/traffic analysis
Nikto	Web-server security assessment
OWASP ZAP	Web-application security testing
Python	Dataset analysis and visualization
Kali Linux utilities	Security assessment and analysis

8. Vulnerability/Risk Register

This is an **important part** of your assignment.

ID	Asset	Vulnerability	Likelihood	Impact	Risk	Recommendation
V01	IoMT device	Unnecessary exposed service	Medium	High	High	Restrict service
V02	Web server	Outdated software	Medium	High	High	Apply security patches
V03	Wireless network	Weak configuration	Medium	High	High	Use secure authentication/encryption
V04	Linux server	Excessive open ports	Medium	Medium	Medium	Close unnecessary ports
V05	Windows workstation	Missing security updates	Medium	High	High	Apply patches
V06	EHR portal	Web security weakness	Medium	High	High	Correct configuration/application weakness

9. Validation Metrics

The assignment specifically asks for measurable validation.

You should include:

1. Exposed Services

Before Security Controls → After Security Controls

2. Attack Detection Rate

$$\text{Detection Rate} = \frac{\text{Detected Attacks}}{\text{Total Attacks}} \times 100$$

3. False Alarm Rate

$$\text{False Alarm Rate} = \frac{\text{False Positives}}{\text{Total Alerts}} \times 100$$

4. Response Time

Measure the time between:

Attack/Event → Detection → Response

5. Network Availability

$$\text{Availability} = \frac{\text{Available Time}}{\text{Total Time}} \times 100$$

6. Resource Utilization

Measure:

- CPU utilization
- RAM utilization
- Network utilization

10. Before-and-After Validation

This will make your assignment much stronger.

Security Parameter	Before Controls	After Controls
Exposed Services	Actual result	Actual result
Detection Rate	Actual result	Actual result
False Alarm Rate	Actual result	Actual result
Response Time	Actual result	Actual result
Network Availability	Actual result	Actual result
CPU Utilization	Actual result	Actual result
Memory Utilization	Actual result	Actual result

11. Test Cases

Test ID	Test	Expected Result
TC01	Host discovery	Authorized hosts identified
TC02	Port scanning	Open ports identified
TC03	Service enumeration	Services identified
TC04	Packet analysis	Network traffic classified
TC05	Web assessment	Security observations identified
TC06	Windows assessment	Security weaknesses documented
TC07	Linux assessment	Security weaknesses documented
TC08	Wireless assessment	Authorized wireless configuration
TC09	IDS validation	Malicious activity detected
TC10	Post-control scan	Reduced unnecessary exposure

12. Execution Evidence

Your report should contain screenshots of **your actual lab execution**, such as:

1. Kali Linux laboratory setup
2. Network architecture
3. Nmap scan
4. Service enumeration
5. Wireshark traffic analysis
6. Nikto assessment
7. OWASP ZAP assessment
8. Dataset loading
9. Dataset analysis
10. Malicious/normal traffic graphs
11. Firewall configuration
12. IDS alerts
13. Centralized logs
14. Before/after results
15. Final validation dashboard

13. Results and Analysis

The results should answer:

- Which services were exposed?
- Which assets had the highest risk?
- What malicious traffic was identified?
- Which attack behavior was most significant?
- Which security controls reduced exposure?
- How effective was the IDS?
- Did false alarms increase or decrease?
- What was the response time?
- Did the controls affect network availability?
- What was the resource overhead?

14. SDG 9 Mapping

SDG 9 – Industry, Innovation and Infrastructure

This project supports **SDG 9** by designing secure and resilient digital infrastructure for connected healthcare environments. IoMT devices, servers, wireless networks and EHR systems require reliable cybersecurity controls to maintain availability, confidentiality and integrity. Ethical hacking helps identify weaknesses before they can be exploited and supports the development of safer digital infrastructure.

15. Limitations

- Testing is restricted to an isolated authorized environment.
- Dataset traffic may not represent every real-world healthcare network.
- Virtual machines may not perfectly reproduce physical IoMT devices.
- Vulnerability results depend on system configuration.
- Detection systems may generate false positives.
- Real-time healthcare operational constraints are not completely reproduced.

16. Possible Improvements

- Integrate real-time IoMT monitoring.
- Implement machine-learning-based attack detection.
- Add automated vulnerability prioritization.
- Integrate SIEM.
- Implement Zero Trust architecture.
- Add real-time threat intelligence.
- Use stronger device authentication.
- Implement continuous security monitoring.
- Add automated incident response.

17. Individual Contribution

For a 3-member team, you could divide the work like this:

**Memb
er**

Work

Student 1 Architecture design, network segmentation and Nmap assessment

Student 2 CICIoMT2024 analysis, Wireshark analysis and vulnerability assessment

Student 3 Web/wireless assessment, security controls, validation and dashboard

Change this to match the actual work performed.

18. Individual Reflection

Each student needs a **separate one-page reflection**.

Use this structure:

My Contribution

Explain the tasks personally completed.

Design Decisions

Explain why specific security controls/tools were selected.

Challenges

Explain technical difficulties faced during implementation.

Learning

Explain what was learned about:

- Ethical hacking
- Network security
- Vulnerability assessment
- Kali Linux
- IoMT security
- Traffic analysis

SDG Connection

Explain how secure digital infrastructure supports SDG 9.

19. Final Report Structure

I recommend submitting your report in this exact order:

1. Cover Page
2. Certificate/Declaration (if required)
3. Team Members
4. Assignment Title
5. Course Outcomes
6. SDG Mapping
7. Abstract
8. Introduction
9. Problem Statement
10. Problem Formulation
11. Objectives
12. Requirements
13. Constraints and Assumptions
14. Smart Healthcare Architecture
15. Proposed Methodology
16. Module Description
17. CICIoMT2024 Dataset Analysis

18. Ethical Hacking Methodology
19. Nmap Assessment
20. Wireshark Analysis
21. Web Security Assessment
22. Wireless Security Assessment
23. Windows Security Assessment
24. Linux Security Assessment
25. Vulnerability/Risk Register
26. Proposed Security Controls
27. Implementation
28. Test Cases
29. Execution Screenshots
30. Validation Metrics
31. Before/After Comparison
32. Results
33. Analysis
34. Trade-offs and Justification
35. SDG Relevance
36. Limitations
37. Future Improvements

38. Conclusion

39. Individual Contributions

40. Individual Reflection – Student 1

41. Individual Reflection – Student 2

42. Individual Reflection – Student 3

43. References

44. GitHub Repository